# Progress Note: 2-Period Igami Model with Agglomeration

14.273 Mini Project

April 15, 2026

## 1 Overview

This note documents the first extension of Igami (2017) to incorporate **innovation spillovers / agglomeration effects** in a simplified 2-period setting. The goal is to understand whether the presence of other innovative firms raises or lowers the incentive to adopt new technology.

## 2 Model

### 2.1 State and Firm Types

The market state at each period $t$ is $s_t = (n_o, n_b, n_n, n_{pe})$, where:

- $n_o$: old incumbents (have *not* adopted new technology),
- $n_b$: "both" incumbents (have adopted; produce with new tech),
- $n_n$: new entrants (using new technology),
- $n_{pe}$: potential entrants.

Active firms in the product market are those with $n_o + n_b + n_n > 0$. "Both" firms are multi-product: they continue to sell old-gen (at cost $c_o$) while also selling new-gen (at cost $c_{n,t}$). This is the source of the **cannibalization effect**: by expanding new-gen supply, a "both" firm depresses the old-gen price it receives, creating a disincentive to innovate that is absent for pure new entrants.

### 2.2 Timing Within Each Period

1. **Product market:** Cournot competition in *two differentiated markets* (old-generation and new-generation products) with linear inverse demand:

$$P_o = A - \frac{B}{M} Q_o - \frac{\rho B}{M} Q_n, \tag{1}$$

$$P_n = A - \frac{B}{M} Q_n - \frac{\rho B}{M} Q_o, \tag{2}$$

where $\rho \in [0, 1)$ is the cross-market substitution parameter. Old firms ($n_o$) compete only in the old-gen market at cost $c_o$. New entrants ($n_n$) compete only in the new-gen market at cost $c_{n,t}$. "Both" firms ($n_b$) are *multi-product*: they sell in both markets simultaneously and internalize the **cannibalization** effect — their new-gen output reduces demand for their own old-gen product.

2. **Private shocks:** Each firm draws i.i.d. cost shocks $\varepsilon_{i,a} \sim \text{Gumbel}(0, \sigma)$ for each action $a$.

3. **Discrete choices** (sequential order: old → both → new → pe):

   - old: $\{\text{stay}, \text{innovate}, \text{exit}\}$
   - both/new: $\{\text{stay}, \text{exit}\}$
   - pe: $\{\text{enter}, \text{quit}\}$

   Each type observes the *realized* outcomes of all earlier movers before making its own decision.

4. **Transition** to period $t + 1$.

## 2.3 Agglomeration Extension

The new-technology marginal cost depends on the number of firms already using it:

$$c_{n,t} = c_{n,0} - \gamma \cdot (n_{b,t} + n_{n,t}), \qquad \gamma \geq 0. \tag{3}$$

When $\gamma = 0$ the model reduces to the Igami baseline. When $\gamma > 0$, a larger innovative cluster lowers costs for all new-tech firms (knowledge spillover / agglomeration effect).

## 2.4 Value Functions and CCPs

Because firms move sequentially within each period and each type observes the *realized* outcomes of all earlier movers, each type's decision problem is a genuine **single-agent problem**: earlier-mover uncertainty has been resolved, and later-mover choices enter only through pre-computed CCPs. Combined with i.i.d. EVT1 private cost shocks, this yields closed-form logit CCPs at each stage:

$$P(a \mid \tau, \tilde{s}) = \frac{\exp\big(\bar{V}_a(\tau, \tilde{s})/\sigma\big)}{\sum_{a'} \exp\big(\bar{V}_{a'}(\tau, \tilde{s})/\sigma\big)}, \tag{4}$$

where $\tilde{s}$ is the *realized intermediate state* after all earlier movers have acted, and $\bar{V}_a$ is the systematic utility:

$$\bar{V}_a(\tau, \tilde{s}) = \pi_\tau(s_0) + \text{cost}(a) + \beta\, \mathbb{E}[V_{t+1}(\tau', s') \mid \tilde{s},\, a_i = a,\, \text{CCP}_{\text{later}}]\,. \tag{5}$$

Here $\text{CCP}_{\text{later}}$ denotes pre-computed CCPs of same-type peers and all later-moving types; flow profits $\pi_\tau(s_0)$ are evaluated at the period-opening state.

### 2.5 Solution Method

The model is solved by **one-pass backward induction**, following Igami (2017):

1. **Terminal period ($t = 1$):** $V_1(\tau, s_1) = \pi_\tau(s_1)$, where Cournot profits use the agglomeration-adjusted cost $c_{n,1} = c_{n,0} - \gamma(n_{b,1} + n_{n,1})$.

2. **Period $t = 0$:** For each state $s_0$, stages are solved in *reverse move order* (PE $\rightarrow$ new $\rightarrow$ both $\rightarrow$ old), exploiting the sequential structure:

   - **Stage 4 (PE):** Each PE firm observes the full realized intermediate state $\tilde{s}$ and solves a binary logit $\{\text{enter}, \text{quit}\}$. CCPs are pre-computed for all intermediate states reachable from $s_0$.
   - **Stage 3 (new):** Each new firm solves a binary logit $\{\text{stay}, \text{exit}\}$, conditioning on pre-computed PE CCPs to integrate over the PE-stage continuation value.
   - **Stage 2 (both):** Each both firm solves a binary logit, conditioning on new + PE CCPs. Innovators locked in from Stage 1 ($k_{io}$ old firms that chose innovate) are added to $n_b$ before this stage.
   - **Stage 1 (old):** Each old firm solves a multinomial logit $\{\text{stay}, \text{innovate}, \text{exit}\}$, conditioning on all lower-stage CCPs.

   No fixed-point iteration is needed within the period: the sequential structure makes each stage a closed-form single-agent problem given pre-computed downstream CCPs. Marginal CCPs for later-moving types are recovered by integrating over the equilibrium transition distribution (products of independent Binomial/Multinomial draws).

# 3 Calibration

The calibration is illustrative (not estimated from data) and is summarised in Table 1.

Table 1: Illustrative calibration

| Parameter | Description | Value |
|---|---|---|
| $A$ | Demand intercept | 3.0 |
| $B$ | Demand slope | 1.0 |
| $M$ | Market size | 1.0 |
| $c_o$ | Old-tech marginal cost | 1.5 |
| $c_{n,0}$ | New-tech baseline cost | 0.5 |
| $\beta$ | Discount factor | 0.9 |
| $\kappa$ | Innovation cost | 0.3 |
| $\phi$ | Entry cost | 0.2 |
| $\sigma$ | EVT1 scale | 1.0 |
| $\gamma$ | Agglomeration (comparative statics) | varied |
| $\rho$ | Cross-market substitution | 0.5 |

The initial state of interest is $s_0 = (n_o, n_b, n_n, n_{pe}) = (4, 1, 1, 2)$, loosely inspired by the early HDD market structure in Igami (2017).

# 4 Results

## 4.1 Baseline vs. Agglomeration

Table 2 compares equilibrium CCPs at $s_0$ for $\gamma = 0$ (baseline) and $\gamma = 0.05$.

Table 2: CCPs at $s_0 = (4, 1, 1, 2)$: baseline vs. agglomeration ($\gamma = 0.05$)

| | $\gamma = 0$ | $\gamma = 0.05$ |
|---|---|---|
| $P(\text{innovate} \mid \text{old})$ | 0.3135 | 0.3206 |
| $P(\text{enter} \mid \text{pe})$ | 0.5061 | 0.5139 |

Agglomeration raises both the innovation and entry rates. Intuitively, a larger innovative cluster lowers future costs for new-tech firms, increasing the expected continuation value of innovating or entering.

## 4.2 Comparative Statics in $\gamma$

Table 3 reports equilibrium CCPs as $\gamma$ varies from 0 to 0.30, and Figure 1 plots the same.

Table 3: CCPs vs. agglomeration parameter $\gamma$

| $\gamma$ | $P(\text{innovate}\|\text{old})$ | $P(\text{enter}\|\text{pe})$ |
|---|---|---|
| 0.00 | 0.3135 | 0.5061 |
| 0.05 | 0.3206 | 0.5139 |
| 0.10 | 0.3282 | 0.5221 |
| 0.15 | 0.3314 | 0.5258 |
| 0.20 | 0.3318 | 0.5263 |
| 0.25 | 0.3315 | 0.5260 |
| 0.30 | 0.3316 | 0.5260 |

## 4.3 Comparative Statics in $\rho$ (Cannibalization)

Figure 2 plots equilibrium CCPs as the substitution parameter $\rho$ varies from 0 (independent markets) to 0.95 (near-homogeneous goods), holding $\gamma = 0.05$.

## 4.4 Discussion

Three features of the comparative statics are notable:

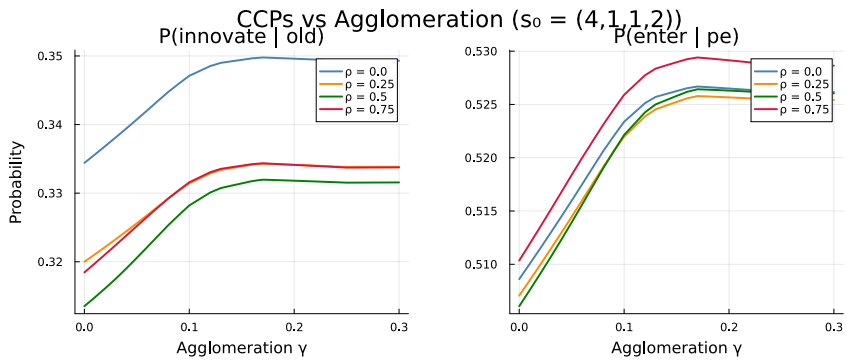


Figure 1: $P(\text{innovate} \mid \text{old})$ and $P(\text{enter} \mid \text{pe})$ as functions of $\gamma$ at $s_0 = (4, 1, 1, 2)$.

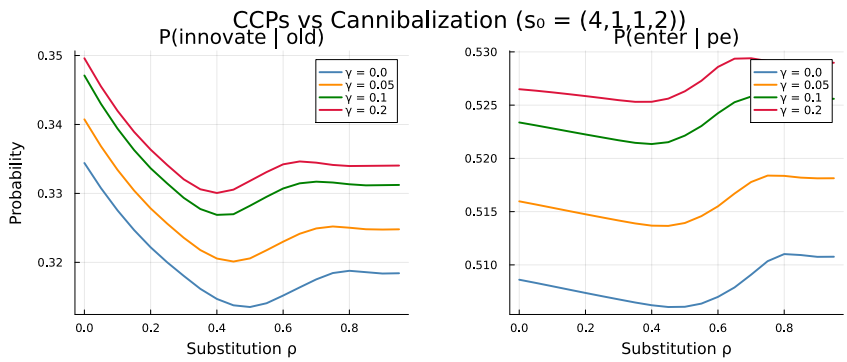


Figure 2: $P(\text{innovate} \mid \text{old})$ and $P(\text{enter} \mid \text{pe})$ as functions of $\rho$ at $s_0 = (4, 1, 1, 2)$, $\gamma = 0.05$.

1. **Positive agglomeration effect:** Higher $\gamma$ unambiguously raises the innovation rate for small values of $\gamma$. This confirms the model's central prediction: knowledge spillovers encourage adoption.

2. **Cannibalization deters innovation:** Higher $\rho$ (stronger substitution between old-gen and new-gen) reduces the profit of "both" firms by depressing old-gen prices. In the calibration, per-firm "both" profit falls from $\approx 0.531$ at $\rho = 0$ to $\approx 0.327$ at $\rho = 0.5$. This is the central mechanism in Igami (2017): incumbents are reluctant to innovate because doing so cannibalizes their existing old-gen revenue.

3. **Saturation:** The innovation rate saturates near $1/3$ for large $\gamma$. This is a feature of the EVT1 / logit structure: with large $\sigma$ relative to payoff differences, all actions approach equal probability. In a richer calibration (smaller $\sigma$), the effect would be more pronounced.

# 5 Regional Extension and Counterfactuals

## 5.1 Regional Structure

To study the interaction between industrial policy and agglomeration we extend the state space with a **region** attribute. There are $R = 3$ regions (labelled $r \in \{1, 2, 3\}$); each firm is born into a region and never leaves it, and potential entrants sit in per-region pools $n_{pe,r}$. Product-market competition remains **global** — all active firms meet in the same two-market Cournot game of equations (1)–(2) — but agglomeration spillovers are now **local**:

$$c_{n,r,t} = c_{n,0} - \gamma_r \cdot \big(n_{b,r,t} + n_{n,r,t}\big), \qquad r = 1, \ldots, R. \tag{6}$$

Because $c_{n,r,t}$ varies across regions, firms of type "both" and "new" face *heterogeneous* marginal costs in the new-gen market. The Cournot stage therefore loses symmetry across regions, and equilibrium quantities are obtained by solving an asymmetric FOC system of up to $4R$ equations per state. Within-region slots remain symmetric (all firms in the same $(type, r)$ cell pick the same quantity), so per-firm profits are still characterised by $\pi_{o,r} = bq^2_{oo,r}$, $\pi_{n,r} = bq^2_{nn,r}$, and $\pi_{b,r} = b(q^2_{bo,r} + q^2_{bn,r}) + 2s\, q_{bo,r} q_{bn,r}$ where $b = B/M$ and $s = \rho B/M$.

Each firm privately captures the lower cost when it innovates but does not internalize the benefit it confers on regional peers — a standard **local agglomeration externality** that creates a wedge between the social and private return to innovation and opens the door to regionally targeted industrial policy.

## 5.2 Solution with Regional States

The sequential-move structure (old → both → new → pe) carries over unchanged, and we extend the same idea one level down: **within each stage, regions move sequentially**, $r = 1 \to 2 \to 3$. When firms in region $r$ decide, they observe the realized outcome of regions $1, \ldots, r-1$ and integrate over the (yet-to-play) strategies of regions $r+1, \ldots, R$. This is the regional analogue of Igami's cross-type sequencing: it makes every sub-stage a single-agent problem whose CCP is a *scalar* fixed point rather than a joint fixed point across regions, and it removes any coupling in the solver between regions at the same stage.

Concretely, we solve each stage in reverse region order. Region $R$'s scalar FP is computed first for every reachable sub-state, and cached. Region $R-1$'s FP is then computed given the cached region-$R$ continuation, and so on down to region 1. Across stages, the usual layered caches still apply: stage-4 (PE) depends only on the post-new-stage state and is reused across initial states via $\mathrm{EV}_{\text{after pe}}[s_{an}, r]$; stages 3, 2, 1 cascade through $\mathrm{EV}_{\text{after new}}$, $\mathrm{EV}_{\text{after both}}$, and $\mathrm{EV}_{\text{after old}}$, each keyed by $(\text{State}, r)$. Because each sub-state's CCP is a function of the realized history, the per-region CCPs reported in the results below are *marginalised* over the equilibrium play of earlier-moving regions, computed by a forward traversal of the sub-stage tree.

Because the state space now has $4R = 12$ integer coordinates, we cap $N_{\max} = 6$ on this branch (about $1.8 \times 10^4$ enumerated states) and pick initial states with total firm count $\leq 6$. One visible artefact of the sequential-move convention is a small first-mover asymmetry in the baseline: with uniform $\gamma_r = 0.05$ the reported innovation CCPs are $(0.3333, 0.3330, 0.3326)$ rather than the exactly-symmetric values one would get under simultaneous moves within

a stage. Region 1 commits without observing anyone, while region 3 best-responds to the realised history; the gap is $\leq 10^{-3}$ in all experiments and does not affect any of the counterfactual comparisons.

## 5.3 Baseline and Counterfactuals

We solve the regional model at two initial states, both with total firm count 6:

- **Symmetric $s_0$:** $s_0^{\text{sym}} = \big((1,1,1),(1,1,1),(0,0,0),(0,0,0)\big)$ — one old incumbent and one "both" incumbent in each region, no entrants.
- **Clustered $s_0$:** $s_0^{\text{clust}} = \big((1,1,1),(3,0,0),(0,0,0),(0,0,0)\big)$ — same number of firms of each type, but all three "both" incumbents are concentrated in region 1.

Starting from $s_0^{\text{sym}}$ with uniform $\gamma_r = 0.05$ we verify a **symmetry check**: the three regions' CCPs are nearly identical, with only tiny differences (on the order of $10^{-3}$) that reflect the sequential region-move structure discussed above rather than any substantive break in the model's permutation symmetry when the state and parameters are symmetric.

Two counterfactuals are then run against this baseline. **Counterfactual A** subsidises innovation in region 1 by tripling its agglomeration strength to $\gamma_1 = 0.15$ while leaving $\gamma_2 = \gamma_3 = 0.05$. **Counterfactual B** holds $\gamma_r$ uniform but starts from the clustered initial state $s_0^{\text{clust}}$, asking whether a historical concentration of innovators is enough to lock in innovation leadership even without a policy tilt.

Table 4: Regional innovation CCPs $P(\text{innovate} \mid \text{old}, r)$ at each scenario. Baseline uses $s_0^{\text{sym}}$ with uniform $\gamma = 0.05$; CF-A sets $\gamma_1 = 0.15$; CF-B uses $s_0^{\text{clust}}$.

| Scenario | $P(\text{innov}\|o, r{=}1)$ | $P(\text{innov}\|o, r{=}2)$ | $P(\text{innov}\|o, r{=}3)$ |
|---|---|---|---|
| Baseline | 0.3333 | 0.3330 | 0.3326 |
| CF-A: subsidy r1 | 0.3594 | 0.3263 | 0.3260 |
| CF-B: cluster r1 | 0.3374 | 0.3236 | 0.3234 |

## 5.4 Discussion

Three findings stand out in Table 4 and Figure 3:

1. **Targeted subsidies work, but leak.** Raising $\gamma_1$ from 0.05 to 0.15 moves region 1's innovation probability from 0.3333 to 0.3594, a 0.3333-to-0.3594 gain of roughly two and a half percentage points. Regions 2 and 3 fall from 0.3330 to 0.3263— a small but clearly negative spillover driven entirely by the *global* product market: more innovation in region 1 lowers expected new-gen prices everywhere, eroding the option value of innovating in the other regions.
2. **Historical clusters alone are weak.** Starting from $s_0^{\text{clust}}$ the incumbent cluster in region 1 barely moves innovation there (0.3374 vs. 0.3333 in the symmetric baseline): with only one remaining old firm in region 1, the region's marginal old firm is already

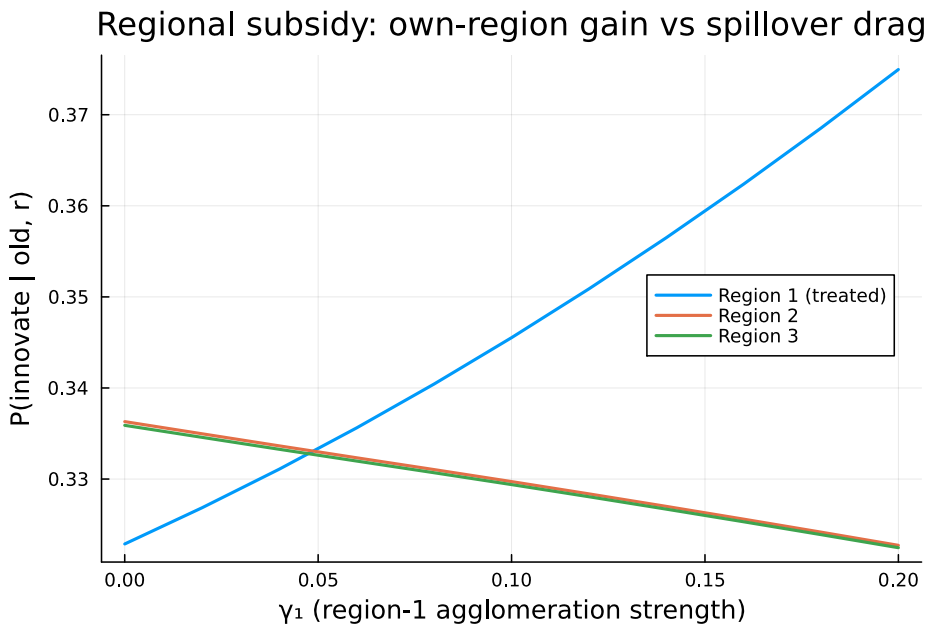


Figure 3: Regional innovation rates as $\gamma_1$ varies while $\gamma_2 = \gamma_3 = 0.05$ is held fixed. Region 1 responds positively to its own subsidy; regions 2 and 3 are slightly depressed through the global Cournot channel.

close to saturating its own local spillover. The larger effect shows up in regions 2 and 3, whose CCPs fall to 0.3236 because of the global-market pressure from the concentrated region-1 innovators. A one-shot historical endowment is therefore a much weaker policy lever than a continuing subsidy.

3. **The spillover wedge is still local.** Comparing CF-A and CF-B makes clear that what drives region 1's innovation is not how many innovators are in the region today but the *marginal reduction in new-tech cost* each additional innovator delivers. The policy CF-A tilts that reduction directly, which is why it dominates a pure history-based CF-B at the same total cluster size.

The two experiments illustrate how the regional model bridges Igami's replacement-effect story with the industrial-policy literature: even a modest, regionally targeted spillover subsidy reshapes the geographic distribution of innovation at the cost of a small drag on the rest of the world, with magnitudes that depend on the level of cannibalization $\rho$ and the strength of the local externality $\gamma_r$.

# 6 Estimation on Simulated Data

As a first pass at recovering the model's structural parameters from data, we simulate a firm-level dataset from the regional model under the calibration of Table 1 and then estimate $(\gamma, \kappa, \phi)$, treating the demand parameters, the EVT1 scale $\sigma$, the cost $c_{n,0}$, and the state cap $N_{\max}$ as known.

## 6.1 Simulated dataset

The simulator (`code/scripts/simulate_data.jl`) draws 500 independent markets from the 2-period regional model. Each market gets its own initial state $s_0 \sim \text{unif}\{0,1,2\}^{4R}$ (rejecting empty or over-capacity draws), so the panel covers a range of local cluster sizes. For every firm in every period we record region, firm type, the two Cournot quantities, realised profit, and—for period 1 only—the discrete action. The structural parameters used to draw the data are deliberately *not* stored in the CSV.

## 6.2 Two-step estimator

Because equilibrium quantities are a deterministic function of the state and the spillover vector $\gamma$, identification splits cleanly: $\gamma$ is pinned by the Cournot block and $(\kappa, \phi)$ by the discrete actions, conditional on $\gamma$. We exploit this with a two-step estimator (see `notes/estimation.md` for the full writeup).

**Step 1 — NLS for $\gamma$.** For each market we reconstruct $s_0$ and $s_1$ by counting firms of each type and region in the corresponding period. We then solve

$$\widehat{\gamma} \;=\; \arg\min_{\gamma} \sum_{m,t,r,\,\text{slot}} \big(q^{\text{obs}}_{m,t,r,\text{slot}} - \hat{q}_{m,t,r,\text{slot}}(\gamma)\big)^2, \tag{7}$$

starting from $\gamma^0 = (0,0,0)$ and leaving $\gamma$ *unconstrained*. Because the simulated quantities are noiseless, the minimum is machine zero (SSR $= 1.34e-13$) and $\widehat{\gamma}$ equals the truth to all reported digits; this step doubles as a sanity check that state reconstruction is correct.

**Step 2 — MLE for $(\kappa, \phi)$.** Holding $\gamma = \widehat{\gamma}$ fixed, we maximise the period-1 action log-likelihood over $(\kappa, \phi)$. The likelihood *replays* the sequential-move order used by the solver: within each stage, regions act in order $r = 1 \to 2 \to 3$, and each firm's contribution uses the stage CCP evaluated at the sub-state its firm actually faced (not the marginal CCP integrated over the forward traversal). Per market,

$$\begin{aligned}
\ell_m(\kappa,\phi) = &\sum_{r=1}^{R}\Big[k_r^{\text{so}}\log p_{\text{so}}(\tilde{s}_r^{\text{OLD}}) + k_r^{\text{io}}\log p_{\text{io}}(\tilde{s}_r^{\text{OLD}}) + k_r^{\text{eo}}\log\big(1-p_{\text{so}}-p_{\text{io}}\big)\Big] \\
&+ \sum_{r=1}^{R}\Big[k_r^{\text{sb}}\log p_{\text{sb}}(\tilde{s}_r^{\text{BOTH}}) + k_r^{\text{eb}}\log\big(1-p_{\text{sb}}\big)\Big] \\
&+ \sum_{r=1}^{R}\Big[k_r^{\text{sn}}\log p_{\text{sn}}(\tilde{s}_r^{\text{NEW}}) + k_r^{\text{en}}\log\big(1-p_{\text{sn}}\big)\Big] \\
&+ \sum_{r=1}^{R}\Big[k_r^{\text{ep}}\log p_{\text{ep}}(\tilde{s}_r^{\text{PE}}) + k_r^{\text{op}}\log\big(1-p_{\text{ep}}\big)\Big],
\end{aligned} \tag{8}$$

where the $k_r^{\bullet}$ are observed action counts in region $r$ and the $\tilde{s}_r^{\bullet}$ are the sub-states produced by sequentially updating $s_0$ with the realised outcomes of earlier regions and stages.

The total objective is $\ell(\kappa, \phi) = \sum_m \ell_m(\kappa, \phi)$, maximised by Nelder–Mead starting from $(\kappa^0, \phi^0) = (1.0, 1.0)$ — intentionally far from the calibration so the optimiser must traverse the parameter space. Standard errors come from the BHHH estimator $V = (\sum_m g_m g_m')^{-1}$ with central finite-difference gradients.

### 6.3 Results

Table 5 reports the estimates against the data-generating parameters. The two-step estimator recovers $\widehat{\gamma} = (0.0500, 0.0500, 0.0500)$ exactly (Step 1), and the action MLE delivers $\widehat{\kappa} = 0.2849$ (s.e. 0.0815, truth 0.3000) and $\widehat{\phi} = 0.1635$ (s.e. 0.0792, truth 0.2000). Both $\kappa$ and $\phi$ lie within roughly one standard error of their true values; given 500 markets and a handful of deciders per market, this is about the precision the sample supports. The period-1 action log-likelihood at the optimum is $-2079.207$.

Table 5: Two-step estimation of $(\gamma, \kappa, \phi)$ on 500 simulated markets. Step 1 identifies $\gamma$ from Cournot quantities; Step 2 estimates $(\kappa, \phi)$ by period-1 action MLE conditional on $\widehat{\gamma}$. Standard errors (BHHH) are reported only for the MLE step.

| Parameter | Truth | Estimate | S.E. |
|---|---|---|---|
| $\kappa$ | 0.3000 | 0.2849 | 0.0815 |
| $\phi$ | 0.2000 | 0.1635 | 0.0792 |
| $\gamma_1$ | 0.0500 | 0.0500 | — |
| $\gamma_2$ | 0.0500 | 0.0500 | — |
| $\gamma_3$ | 0.0500 | 0.0500 | — |

## 7 Next Steps

1. Replace the illustrative calibration with Igami's (2017) structural estimates (Table 3 of the paper).
2. Extend to $T > 2$ periods (the HDD panel spans 1977–1999).
3. Richer counterfactuals on the regional branch: bilateral spillover pools (e.g. regions 1–2 share an enlarged effective $N^{\text{inn}}$), endogenous region choice at entry, and a welfare accounting that nets consumer surplus against the fiscal cost of the subsidy.
4. Study welfare and industrial policy implications (e.g., R&D subsidies).